TOPIC #

The Gram Schmidt Process:

$\Rightarrow$ This process is used to convert given basis into orthonormal basis.

$\Rightarrow$ Conditions for orthonormal basis:

1) All vectors should be perpendicular $\langle u_i, u_j \rangle =$ if $i \neq j$
2) $\langle u_i, u_i \rangle = 1$

$\Rightarrow$ Given a basis $\{v_1, v_2, \dots, v_n\}$ of inner product space V over R.
An orthonormal basis $\{u_1, u_2, u_3, \dots, u_n\}$ of V can be constructed as follows.

Step 1: $u_1 = \dfrac{v_1}{\|v_1\|}$

Step 2:

$$w_2 = v_2 - \langle v_2, u_1 \rangle u_1$$

and $u_2 = \dfrac{w_2}{\|w_2\|}$

Step 3:

$$w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$$

and $u_3 = \dfrac{w_3}{\|w_3\|}$

$\vdots$

and so on

$w_n = v_n - \langle v_n, u_1 \rangle u_1 - \langle v_n, u_2 \rangle u_2$
$- \langle v_n, u_3 \rangle u_3 \cdots - \langle v_n, u_{n-1} \rangle u_{n-1}$

$$u_n = \frac{w_n}{\|w_n\|}$$

Show that $\{(1,1,1), (0,1,1), (0,0,1)\}$ is a basis of R^3. Using gram schmidt orthonormalization process to transform into orthonormal basis (PP-2021, 2020)

Let

$S = \{ (1,1,1), (0,1,1), (0,0,1)\}$

$v_1 = (1,1,1)$

$v_2 = (0,1,1)$

$v_3 = (0,0,1)$

<u>Check Linear independence:</u>

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 0 + 0$$
$$= 1 \neq 0$$

⟹ The given matrix is non-singular and is in echelon form, Therefore given vectors are linearly independent and so form a basis of R_3

Step 1:

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{(1,1,1)}{\sqrt{(1)^2+(1)^2+(1)^2}}$$

$$= \frac{(1,1,1)}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

Step 2:

$$w_2 = v_2 - \langle v_2, u_1 \rangle u_1$$

$$w_2 = (0,1,1) - \left((0,1,1)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)\right)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \left(0 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \left(0 + \frac{2}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$w_2 = \left(\frac{-2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$u_2 = \frac{(-2/3, 1/3, 1/3)}{\sqrt{\frac{4}{9}+\frac{1}{9}+\frac{1}{9}}} = \frac{(-2/3, 1/3, 1/3)}{\sqrt{6/9}}$$

$= \frac{-2/3}{\sqrt{6}/3}, \frac{1/3}{\sqrt{6}/3}, \frac{1/3}{\sqrt{6}/3} = \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

Step3

$w_3 = V_3 - \langle V_3, u_1\rangle u_1 - \langle V_3, u_2\rangle u_2$

$= (0,0,1) - \langle (0,0,1)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)\rangle \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$- \langle (0,0,1), \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)\rangle \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

$= (0,0,1) - \langle 0 + 0 + \frac{1}{\sqrt{3}}\rangle \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$- \langle (0,0,1)\left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)\rangle \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

$= (0,0,1) - \left(\frac{1}{\sqrt{3}}\right)\left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

$- \left(0 + 0 + \frac{1}{\sqrt{6}}\right)\left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

$= (0,0,1) - \left(\frac{-2}{\sqrt{18}}, \frac{1}{3}, \frac{1}{3}\right)$

$- \left(\frac{-2}{6}, \frac{1}{6}, \frac{1}{6}\right)$

$= \left(0 - \frac{1}{3} + \frac{2}{6}, 0 - \frac{1}{3} - \frac{1}{6}, 1 - \frac{1}{3} - \frac{1}{6}\right)$

$w_3 = (0, -1/2, -1/2)$

$$u_3 = \frac{w_3}{\|w_3\|} = \frac{(0, -1/2, -1/2)}{\sqrt{0 + 1/4 + 1/4}}$$

$$= \frac{(0, -1/2, -1/2)}{\sqrt{1/2}}$$

$$= \left(0, \frac{-1}{2}/\frac{1}{\sqrt{2}}, \frac{-1}{2}/\frac{1}{\sqrt{2}}\right)$$

$$= \left(0, \frac{-1}{2} \times \sqrt{2}, \frac{-1}{2} \times \sqrt{2}\right)$$

$$= \left(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

Therefore orthogonal basis.

$$\left\{\left(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}\right), \left(-2/\sqrt{6}, 1/\sqrt{6}, 1/\sqrt{6}\right), \left(0, -1/\sqrt{2}, -1/\sqrt{2}\right)\right\}$$

Show that $u = (-2, 3, 1, 4)$ and $V = (1, 2, 0, -1)$ are orthogonal vectors in R. (PP-2019)

∴ Two vectors are said to be orthogonal if

$$\langle u \cdot v \rangle = 0$$

$$= \langle (-2, 3, 1, 4) \cdot (1, 2, 0, -1) \rangle$$

$$= (-2)(1) + (3)(2) + (1)(0) + (4)(-1)$$

$$= -2 + 6 + 0 - 4$$

$$= 4 - 4$$

$$= 0$$

Cross product:

=> if $u = (u_1, u_2, u_3)$ and $V = (V_1, V_2, V_3)$ are vectors in 3-space, then the cross product $u \times V$ is the vector defined by

$$u \times V = (u_2 V_3 - u_3 V_2, u_3 V_1 - u_1 V_3, u_1 V_2 - u_2 V_1)$$

or in determinant notation

$$u \times V = \left(\begin{vmatrix} u_2 & u_3 \\ V_2 & V_3 \end{vmatrix} - \begin{vmatrix} u_1 & u_3 \\ V_1 & V_3 \end{vmatrix} \begin{vmatrix} u_1 & u_2 \\ V_1 & V_2 \end{vmatrix} \right)$$

=> Also called scalar product

Find a unit vector perpendicular.

$\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ & $\vec{b} = -2\hat{i} + \hat{j} - 3\hat{k}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ -2 & 1 & -3 \end{vmatrix} = \hat{i}(-3-2) - \hat{j}(-3+4) + \hat{k}(1+2)$$

$$= -5\hat{i} - \hat{j} + 3\hat{k}$$

$$\hat{V} = -5\hat{i} - \hat{j} + 3\hat{k}$$

$$\hat{V} = \frac{\vec{V}}{|V|} = \frac{(-5\hat{i} - \hat{j} + 3\hat{k})}{\sqrt{(-6)^2 + (-1)^2 + (3)^2}}$$

$$= \frac{-5\hat{i} - \hat{j} + 3\hat{k}}{\sqrt{90}}$$

The Gram-Schmidt Process

V is an I.P. Space over 'R'.

$\{v_1, v_2 \ldots v_n\}$ is a basis of $V(R)$.

An orthonormal basis $\{u_1, u_2 \ldots u_n\}$ of V can be constructed as,

Step 1 Let $u_1 = v_1 \cdot \frac{1}{\|v_1\|}$

$$\|u_1\| = \|v_1\| \cdot \frac{1}{\|v_1\|} = 1$$

u_1 is written in the linear combination of v_1 & $\|v_1\|$ where $\|v_1\|$ is scalar.
Similarly $v_1 = u_1 \|v_1\|$
So $[u_1] = [v_1]$

Step 2

Let $u_2 = \frac{w_2}{\|w_2\|}$ where $w_2 = v_2 - \langle v_2, u_1\rangle u_1$

$$\langle u_1, u_2\rangle = \langle u_1, \frac{w_2}{\|w_2\|}\rangle$$

$$= \frac{1}{\|w_2\|}\langle u_1, w_2\rangle$$

$$= \frac{1}{\|w_2\|}\langle u_1, v_2 - \langle v_2, u_1\rangle u_1\rangle$$

$$= \frac{1}{\|w_2\|}\left[\langle u_1, v_2\rangle - \langle v_2, u_1\rangle\langle u_1, u_1\rangle\right]$$ by cond iii

$$= \frac{1}{\|w_2\|}\left[\langle u_1, v_2\rangle - \langle u_1, v_2\rangle \cdot 1\right]$$

$\because \|u_1\| = \sqrt{\langle u_1, u_1\rangle}$
$1 = \sqrt{\langle u_1, u_1\rangle}$
squaring $1 = \langle u_1, u_1\rangle$

$\langle u_1, u_2\rangle = \boxed{0}$ & $\|u_2\| = \frac{\|w_2\|}{\|w_2\|} = 1$

Hence $[u_1, u_2]$ is orthonormal

Hence $[u_1, u_2]$ is O.N. syst.

Step 3 Let $u_3 = \frac{w_3}{\|w_3\|}$ where $w_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$

Similarly $[u_1, u_2, u_3]$ is orthonormal.

and so on, then $u_n = \frac{w_n}{\|w_n\|}$ where $w_n = v_n - \langle v_n, u_1\rangle u_1 - \langle v_n, u_2\rangle u_2 \ldots \ldots - \langle v_n, u_{n-1}\rangle u_{n-1}$

$\therefore$ The set $\{u_1, u_2, \ldots u_n\}$ is orthonormal

$\because$ We know orthonormal is linearly independent and Span $\{u_1, u_2, \ldots u_n\}$ = Span $\{v_1, v_2, \ldots v_n\}$

Hence $\{u_1, u_2, \ldots u_n\}$ is orthonormal basis of V.

Theorem Every orthonormal system $\{u_1, u_2, \ldots u_n\}$ is linearly independent. Moreover, for all $v \in V$, the vector $w = v - \sum_{K=1}^{n} \langle v, u_K \rangle u_K$ is orthogonal to each u_i, $1 \le i \le n$. ie $\{u_1, u_2, \ldots u_n\}$

Proof Suppose $a_1 u_1 + a_2 u_2 + \cdots a_n u_n = 0$ where a_i are scalars $a_i \in R$

Taking innerproduct of both sides with 'u_i'

$$\langle a_1 u_1 + a_2 u_2 \cdots + a_n u_n, u_i \rangle = \langle 0, u_i \rangle$$

$$a_1 \langle u_1, u_i \rangle + a_2 \langle u_2, u_i \rangle + \cdots a_{i-1} \langle u_{i-1}, u_i \rangle$$

$$+ a_i \langle u_i, u_i \rangle + a_{i+1} \langle u_{i+1}, u_i \rangle + \cdots a_n \langle u_n, u_i \rangle = 0$$

$\because$ Cond iii

$\because 0$ is orthogonal to every $v \in V$ $\therefore \langle 0, v \rangle = 0$

$$0 + 0 + \cdots 0 + a_i \cdot 1 + 0 + \cdots 0 = 0$$

$$a_i \cdot 1 = 0$$

$$a_i = 0 \quad \forall i = 1, 2, \ldots n$$

$\because 1 \ne 0$ $a_i = 0$

$\because \{u_1, u_2 \ldots u_n\}$ is orthonormal system $\therefore \langle u_i, u_j \rangle = 0$, $i \ne j$ Distinct; $\langle u_i, u_i \rangle = 1$ $i = j$

$\therefore \{u_1, u_2 \ldots u_n\}$ is linearly independent.

Now to prove w is orthogonal to each u_i, $1 \le i \le n$

Consider

$$\langle w, u_i \rangle = \left\langle v - \sum_{K=1}^{n} \langle v, u_K \rangle u_K, u_i \right\rangle$$

$$= \langle v, u_i \rangle - \left\langle \sum_{K=1}^{n} \langle v, u_K \rangle u_K, u_i \right\rangle \quad \because \text{Cond iii}$$

$$= \langle v, u_i \rangle - \langle \langle v, u_1 \rangle u_1 + \langle v, u_2 \rangle u_2, \ldots \langle v, u_n \rangle u_n, u_i \rangle$$

$$= \langle v, u_i \rangle - (\langle v, u_1 \rangle) \langle u_1, u_i \rangle - (\langle v, u_2 \rangle) \langle u_2, u_i \rangle, \ldots$$

$$- (\langle v, u_{i-1} \rangle) \langle u_{i-1}, u_i \rangle - \langle v, u_i \rangle \langle u_i, u_i \rangle$$

$\because \langle v, u \rangle \in F = R$ $\therefore$ by Cond iii

$$- (\langle v, u_{i+1} \rangle) \langle u_{i+1}, u_i \rangle \cdots - (\langle v, u_n \rangle) \langle u_n, u_i \rangle$$

$$= \langle v, u_i \rangle - 0 - 0 \cdots - 0 - \langle v, u_i \rangle \cdot 1 - 0 \cdots 0$$

$\left(\because \langle u_i, u_j \rangle = 0 \right.$, $\langle u_i, u_i \rangle = 1$

$$= \langle v, u_i \rangle - \langle v, u_i \rangle$$

$$\langle w, u_i \rangle = \boxed{0}$$

Hence w is orthogonal to each u_i, $1 \le i \le n$

Example 12 Show that $\{(1,1,1),(0,1,1),(0,0,1)\}$ is a basis of R^3. Using Gram-Schmidt orthogonalization process, transform this basis into an orthonormal basis.

Sol. Let $(x,y,z) \in R^3$

Suppose $(x,y,z) = a(1,1,1) + b(0,1,1) + c(0,0,1)$ — Linear Combination of a,b,c & given set of vectors

$(x,y,z) = (a,\ a+b,\ a+b+c)$

$\therefore \boxed{a = x}$

$a+b = y \quad \Rightarrow b = y-a \quad \Rightarrow \boxed{b = y-x}$

$a+b+c = z \quad \Rightarrow c = z-a-b \quad \Rightarrow c = z-x-(y-x)$

$c = z-x-y+x$

$\boxed{c = z-y}$

$\therefore$ the given set of vectors $\{(1,1,1),(0,1,1),(0,0,1)\}$ can be written as a linear combination of a,b,c, as values of a,b,c exist.

$\therefore$ the set $\{(1,1,1),(0,1,1),(0,0,1)\}$ is a spanning set for R^3

<u>Now check Linear Independence</u> of $\{(1,1,1),(0,1,1),(0,0,1)\}$

$a(1,1,1) + b(0,1,1) + c(0,0,1) = 0$

$(a,\ a+b,\ a+b+c) = 0$

$\boxed{a = 0}$

$a+b = 0 \quad \Rightarrow 0+b = 0 \quad \Rightarrow \boxed{b=0}$

$a+b+c = 0 \quad \Rightarrow 0+0+c = 0 \quad \Rightarrow \boxed{c=0}$

Hence given vectors are Lin. Independent.

<u>2nd Method</u>

Since the Matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Rank of $A = 3$ = No. of vectors $= 3$

is in Echelon form

So the given three vectors $(1,1,1),(0,1,1),(0,0,1)$ are Linearly Independent.

Since the given set of vectors is a Spanning Set for R^3 and is Linearly Independent. Hence

Hence the given set of vectors $\{(1,1,1),(0,1,1),(0,0,1)\}$ is a basis of R^3.

P.T.O.

By Gram Schmidt Orthogonalisation:

Let $v_1 = (1,1,1)$, $v_2 = (0,1,1)$, $v_3 = (0,0,1)$

Now $u_1 = \frac{v_1}{\|v_1\|} = \frac{(1,1,1)}{\sqrt{1^2+1^2+1^2}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

$u_2 = \frac{w_2}{\|w_2\|}$ where $w_2 = v_2 - \langle v_2, u_1\rangle u_1$

$$= (0,1,1) - \left\langle (0,1,1), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)\right\rangle \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \left(0 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \frac{2}{\sqrt{3}}\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= (0,1,1) - \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$w_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$\|w_2\| = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{6}{9}} = \sqrt{\frac{2}{3}}$$

$$\therefore u_2 = \frac{\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)}{\sqrt{\frac{2}{3}}}$$

$$u_2 = \sqrt{\frac{3}{2}}\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right) = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$u_3 = \frac{w_3}{\|w_3\|}$ where $w_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$

$$= (0,0,1) - \left\langle (0,0,1), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)\right\rangle \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) -$$

$$- \left\langle (0,0,1), \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)\right\rangle \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= (0,0,1) - \left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) - \left(\frac{1}{\sqrt{6}}\right)\left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= (0,0,1) - \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) - \left(\frac{-2}{6}, \frac{1}{6}, \frac{1}{6}\right)$$

$$= \left(0 - \frac{1}{3} + \frac{2}{6},\ 0 - \frac{1}{3} - \frac{1}{6},\ 1 - \frac{1}{3} - \frac{1}{6}\right)$$

$$= \left(0,\ \frac{-2-1}{6},\ \frac{6-2-1}{6}\right)$$

$$w_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$\|w_3\| = \sqrt{0 + \frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$$

$$\therefore u_3 = \frac{\left(0, -\frac{1}{2}, \frac{1}{2}\right)}{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{2}\left(0, -\frac{1}{2}, \frac{1}{2}\right) = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

Thus the orthonormal basis is $\{u_1, u_2, u_3\} = \{($ [illegible] $, \frac{1}{\sqrt{3}}), (-\frac{2}{\sqrt{6}}$ [illegible]

7. Show that (1, 1) (0, 1) is a basis of R^2. Using the Gram-Schmidt process, find an orthonormal basis of R^2.

8. Show that (1, 0, 1), (0, 1, 1), (0, 0, 1) is a basis of R^3. Using the Gram-Schmidt process, find an orthonormal basis of R^3 by taking

(i) $u_1 = (0, 0, 1)$

(ii) $u_1 = (1, 0, 1)$.

9. Show that $\{(1, -1, 0), (2, -1, -2), (1, -1, -2)\}$ is a basis of R^3. Find an orthonormal basis of R^3 using the Gram-Schmidt process.

10. (i) Let $\{e_1, e_2, \ldots, e_n\}$ be an orthonormal basis of an inner product space V over R. Show that for any $v \in V$,

$$v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \ldots + \langle v, e_n \rangle e_n$$

(ii) If $T : V \to V$ is a linear transformation, show that $\langle T(e_j), e_i \rangle$ is the ij th entry of the matrix representing T in the given basis $\{e_1, e_2, \ldots, e_n\}$.

11. Let W be a subspace of an inner product space V [illegible] that there is an orthonormal basis of W which is part of an orthonormal b[illegible] V.

(i) Show that (1,1) (0,1) is a basis of R^2. Using the Gram-Schmidt process, find an orthonormal basis of R^2.

Sol Let $(x,y) \in R^2$ $\qquad S=\{(1,1),(0,1)\}$

Consider $(x,y) = a(1,1)+b(0,1)$ $\qquad$ linear combination of a, b & given vectors

$(x,y) = (a, a+b)$

$\Rightarrow \boxed{a=x}$

& $a+b=y \Rightarrow x+b=y \Rightarrow \boxed{b=y-x}$

$\therefore$ the linear combination exists, as (1,1) & (0,1) can be written as a linear combination of a & b. So the set 'S' is spanning set for R^2.

Now for Linear Independence.

$a(1,1)+b(0,1)=0$

$\boxed{a=0}$

$a+b=0 \Rightarrow 0+b=0 \Rightarrow \boxed{b=0}$

Hence given vectors are Lin. Independent.

2nd Method

$A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$

It is in echelon form

Rank of A = 2 = No. of vectors in S.

$\therefore$ S is Lin. Independent.

S is spanning set & Linearly Independent So S is basis for R^2

Orthonormal Basis of R^2 by Gram Schmidt Process.

Let $v_1=(1,1)$ $\quad v_2=(0,1)$

Now $u_1 = \dfrac{v_1}{\|v_1\|} = \dfrac{(1,1)}{\sqrt{1^2+1^2}} = \left(\dfrac{1}{\sqrt2}, \dfrac{1}{\sqrt2}\right)$

$u_2 = \dfrac{w_2}{\|w_2\|}$ $\quad$ where $w_2 = v_2 - \langle v_2, u_1\rangle u_1$

$= (0,1) - \langle (0,1), (\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})\rangle (\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})$

$= (0,1) - (0\cdot\tfrac{1}{\sqrt2} + 1\cdot\tfrac{1}{\sqrt2})(\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})$

$= (0,1) - \tfrac{1}{\sqrt2}(\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})$

$= (0,1) - (\tfrac12, \tfrac12)$

$= (0-\tfrac12, 1-\tfrac12)$

$w_2 = (-\tfrac12, \tfrac12)$

$\|w_2\| = \sqrt{(-\tfrac12)^2+(\tfrac12)^2} = \sqrt{\tfrac14+\tfrac14} = \tfrac{1}{\sqrt2}$

$\therefore u_2 = \dfrac{(-\frac12, \frac12)}{\frac{1}{\sqrt2}}$

$u_2 = \sqrt2(-\tfrac12, \tfrac12) = (-\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})$

Thus the orthonormal basis is $\{u_1, u_2\} = \{(\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2}), (-\tfrac{1}{\sqrt2}, \tfrac{1}{\sqrt2})\}$

$\|w_3\| = \sqrt{1^2+0^2+0^2} = 1$

$u_3 = \dfrac{w_3}{\|w_3\|} = \dfrac{(1,0,0)}{1} = (1,0,0)$

Therefore orthonormal basis is $\{u_1, u_2, u_3\}$

Therefore orthonormal basis is $\{u_1, u_2, u_3\} = \{(\ldots), (\ldots), (\ldots)\}$

(ii) Taking $v_1 = (1,0,1)$, $v_2 = (0,1,1)$, $v_3 = (0,0,1)$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{(1,0,1)}{\sqrt{1^2+0^2+1^2}} = \boxed{\left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)}$$

$$u_2 = \frac{\omega_2}{\|\omega_2\|} \quad \text{where } \omega_2 = v_2 - \langle v_2, u_1\rangle u_1$$

$$= (0,1,1) - \left\langle (0,1,1), \left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)\right\rangle \left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)$$

$$= (0,1,1) - \frac{1}{\sqrt2}\left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)$$

$$= (0,1,1) - \left(\frac12, 0, \frac12\right)$$

$$\omega_2 = \left(-\frac12, 1, \frac12\right)$$

$$\|\omega_2\| = \sqrt{\frac14 + 1 + \frac14} = \sqrt{\frac64} = \sqrt{\frac32}$$

$$\therefore u_2 = \frac{\omega_2}{\|\omega_2\|} = \frac{\left(-\frac12, 1, \frac12\right)}{\sqrt{\frac32}} = \frac{\sqrt2}{\sqrt3}\left(-\frac12, 1, \frac12\right) = \left(-\frac{1}{\sqrt6}, \frac{\sqrt2}{\sqrt3}, \frac{1}{\sqrt6}\right)$$

$$= \boxed{\left(-\frac{1}{\sqrt6}, \frac{2}{\sqrt6}, \frac{1}{\sqrt6}\right)}$$

$$u_3 = \frac{\omega_3}{\|\omega_3\|} \quad \text{where } \omega_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$$

$$= (0,0,1) - \left\langle (0,0,1), \left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)\right\rangle \left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right)$$

$$- \left\langle (0,0,1), \left(-\frac{1}{\sqrt6}, \frac{2}{\sqrt6}, \frac{1}{\sqrt6}\right)\right\rangle \left(-\frac{1}{\sqrt6}, \frac{2}{\sqrt6}, \frac{1}{\sqrt6}\right)$$

$$= (0,0,1) - \frac{1}{\sqrt2}\left(\frac{1}{\sqrt2}, 0, \frac{1}{\sqrt2}\right) - \frac{1}{\sqrt6}\left(-\frac{1}{\sqrt6}, \frac{2}{\sqrt6}, \frac{1}{\sqrt6}\right)$$

$$= (0,0,1) - \left(\frac12, 0, \frac12\right) - \left(-\frac16, \frac26, \frac16\right)$$

$$= \left(0-\frac12, 0, \frac12\right) - \left(-\frac16, \frac26, \frac16\right)$$

$$\omega_3 = \left(-\frac12+\frac16, -\frac26, \frac12-\frac16\right) = \left(-\frac13, -\frac13, \frac13\right)$$

$$\|\omega_3\| = \sqrt{\frac19+\frac19+\frac19} = \sqrt{\frac39} = \sqrt{\frac13}$$

$$u_3 = \frac{\omega_3}{\|\omega_3\|} = \frac{\left(-\frac13, -\frac13, \frac13\right)}{\frac{1}{\sqrt3}} = \sqrt3\left(-\frac13, -\frac13, \frac13\right) = \boxed{\left(-\frac{1}{\sqrt3}, -\frac{1}{\sqrt3}, \frac{1}{\sqrt3}\right)}$$

$\{u_1, u_2, u_3\}$ is orthonormal basis.

Show that (1,0,1), (0,1,1), (0,0,1) is a basis of R^3. Using the Gram Schmidt process, find an orthonormal basis of R^3 by taking $u_1=(0,0,1)$

Let $S=\{(1,0,1),(0,1,1),(0,0,1)\} \subset R^3$

Let $(x,y,z) \in R^3$

Consider $(x,y,z) = a(1,0,1)+b(0,1,1)+c(0,0,1)$
$= (a, b, a+b+c)$.

$\boxed{a=x}$
$\boxed{b=y}$
$a+b+c=z \Rightarrow \boxed{c=z-x-y}$ So Linear combination of vectors of S exists and a, b, c. So S is spanning set for R^3.

Now $a(1,0,1)+b(0,1,1)+c(0,0,1)=0$

$a+0+0=0 \Rightarrow \boxed{a=0}$
$0+b+0=0 \Rightarrow \boxed{b=0}$ Hence S is L. independent.
$a+b+c=0 \Rightarrow \boxed{c=0}$

To check Linear Independence. 2nd Method

$A=\begin{bmatrix}1&0&1\\0&1&1\\0&0&1\end{bmatrix}$

Rank of A = 3 = No. of vectors in S.
∴ S is independent.

So S is a basis of R^3 as S is spanning & Lin. independent.

Gram Schmidt Orthonormalization Process.

Taking $v_1=(0,0,1)$ $v_2=(0,1,1)$. $v_3=(1,0,1)$

$u_1 = \frac{v_1}{\|v_1\|} = \frac{(0,0,1)}{\sqrt{0^2+0^2+1^2}} = (0,0,1)$

$u_2 = \frac{\omega_2}{\|\omega_2\|}$ where $\omega_2 = v_2 - \langle v_2, u_1\rangle u_1$
$= (0,1,1) - \langle(0,1,1),(0,0,1)\rangle(0,0,1)$
$= (0,1,1) - (0+0+1)(0,0,1)$
$= (0,1,1) - (0,0,1)$
$\omega_2 = (0,1,0)$
$\|\omega_2\| = \sqrt{0^2+1^2+0^2} = 1$.

$\therefore u_2 = \frac{(0,1,0)}{1} = (0,1,0)$

$u_3 = \frac{\omega_3}{\|\omega_3\|}$ where $\omega_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$
$= (1,0,1) - \langle(1,0,1),(0,0,1)\rangle(0,0,1) - \langle(1,0,1),(0,1,0)\rangle(0,1,0)$
$= (1,0,1) - 1(0,0,1) - 0(0,1,0)$
$= (1,0,1) - (0,0,1)$
$\omega_3 = (1,0,0)$

(9) Show that $\{(1,-1,0),(2,-1,-2),(1,-1,-2)\}$ is a basis of R^3. Find an orthonormal basis of R^3 using Graham Schmidt Process.

Sol $S=\{(1,-1,0),\ (2,-1,-2),\ (1,-1,-2)\}$

Let $(x,y,z)\in R^3$

Consider $(x,y,z)=a(1,-1,0)+b(2,-1,-2)+c(1,-1,-2)$ — (I)

from

$x=a+2b+c$ — (II)

$y=-a-b-c$ — (III)

$z=-2b-2c$ — (IV)

from (II) & (III) $\boxed{x+y=b}$ Put in (IV)

$z=-2x-2y-2c \Rightarrow 2c=-2x-2y-z$

$\boxed{c=-x-y-\frac{z}{2}}$ Put in

from (III) $y=-a-(x+y)-(-x-y-\frac{z}{2})$

$a=-y-\cancel{x}-\cancel{y}+\cancel{x}+\cancel{y}+\frac{z}{2} \Rightarrow \boxed{a=\frac{z}{2}-y}$

Linear combination of a, b, c & vectors of S exist. So S is spaning set.

Now

$a(1,-1,0)+b(2,-1,-2)+c(1,-1,-2)=0$

$a+2b+c=0$ — (V)

$-a-b-c=0$ — (VI)

$-2b+(-2c)=0 \Rightarrow b=\frac{\cancel{2}c}{-\cancel{2}}=\boxed{-c}$ — (VII)

$-a-b-c=0 \Rightarrow -a-(-c)-c=0$

$\Rightarrow -a+c-c=0$

$\Rightarrow \boxed{a=0}$ Put in (V)

$a+2b+c=0 \Rightarrow 0+2(-c)+c=0$

$\Rightarrow \boxed{c=0}$

by Putting $c=0$ in (VII) $\boxed{b=0}$

Hence S is linearly Independent.

$\because$ S is spaning & linearly Independent

So S is basis of R^3.

2nd Method

$$A=\begin{bmatrix}1&-1&0\\2&-1&-2\\1&-1&-2\end{bmatrix}$$

$$\sim\begin{bmatrix}1&-1&0\\0&1&-2\\0&0&-2\end{bmatrix} \text{ Echelon form}$$

Rank of A = 3 = No. of vectors in S.

So S is linearly Independent

Gram Schmidt Process.

$S = \{(1,-1,0),\ (2,-1,-2),\ (1,-1,-2)\}$

$u_1 = \frac{v_1}{\|v_1\|} = \frac{(1,-1,0)}{\sqrt{1^2+1^2+0}} = \left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)$

Let $u_2 = \frac{w_2}{\|w_2\|}$ where $w_2 = v_2 - \langle v_2, u_1\rangle u_1$

$= (2,-1,-2) - \left\langle (2,-1,-2), \left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)\right\rangle \left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)$

$= (2,-1,-2) - \left(\frac{2}{\sqrt2} + \frac{1}{\sqrt2} + 0\right)\left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)$

$= (2,-1,-2) - \frac{3}{\sqrt2}\left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)$

$= (2,-1,-2) - \left(\frac{3}{2}, -\frac{3}{2}, 0\right)$

$w_2 = \left(\frac{1}{2}, +\frac{1}{2}, -2\right)$

$\|w_2\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + (-2)^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \sqrt{\frac{18}{4}}$

$= \sqrt{\frac{9}{2}} = \frac{3}{\sqrt2}$

$\therefore u_2 = \frac{w_2}{\|w_2\|} = \frac{\left(\frac{1}{2}, \frac{1}{2}, -2\right)}{\frac{3}{\sqrt2}} = \frac{\sqrt2}{3}\left(\frac{1}{2}, \frac{1}{2}, -2\right) = \left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right)$

$u_3 = \frac{w_3}{\|w_3\|}$ where $w_3 = v_3 - \langle v_3, u_1\rangle u_1 - \langle v_3, u_2\rangle u_2$

$= (1,-1,-2) - \left\langle (1,-1,-2), \left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right)\right\rangle \left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right) -$

$- \left\langle (1,-1,-2), \left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right)\right\rangle \left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right)$

$= (1,-1,-2) - \left(\frac{1}{\sqrt2} + \frac{1}{\sqrt2} + 0\right)\left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right) - \left(\cancel{\frac{1}{3\sqrt2}} - \cancel{\frac{1}{3\sqrt2}} + \frac{4\sqrt2}{3}\right)\left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right)$

$= (1,-1,-2) - \left(\frac{2}{\sqrt2}\right)\left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right) - \left(\frac{4\sqrt2}{3}\right)\left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right)$

$= (1,-1,-2) - \left(\frac{2}{2}, -\frac{2}{2}, 0\right) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right)$

$= (1-1, -1+1, -2-0) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right)$

$w_3 = (0, 0, -2) - \left(\frac{4}{9}, \frac{4}{9}, -\frac{16}{9}\right) = \left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right)$

$\|w_3\| = \sqrt{\frac{16}{81} + \frac{16}{81} + \frac{4}{81}} = \sqrt{\frac{36}{81}} = \frac{6}{9} = \frac{2}{3}$

$\therefore u_3 = \frac{w_3}{\|w_3\|} = \frac{\left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right)}{2/3} = \frac{3}{2}\left(-\frac{4}{9}, -\frac{4}{9}, -\frac{2}{9}\right) = \left(-\frac{2}{3}, -\frac{2}{3}, -\frac{1}{3}\right)$

$\therefore$ Orthonormal basis is $\{u_1, u_2, u_3\} = \left\{\left(\frac{1}{\sqrt2}, -\frac{1}{\sqrt2}, 0\right), \left(\frac{1}{3\sqrt2}, \frac{1}{3\sqrt2}, -\frac{2\sqrt2}{3}\right), \left(-\frac{2}{3}, -\frac{2}{3}, -\frac{1}{3}\right)\right\}$